ASSIGNMENT REPORT
ON
From BI to Big Data: Explain, Design & Defend

Subject Name: Big Data Analytics
Subject Code: BAD601

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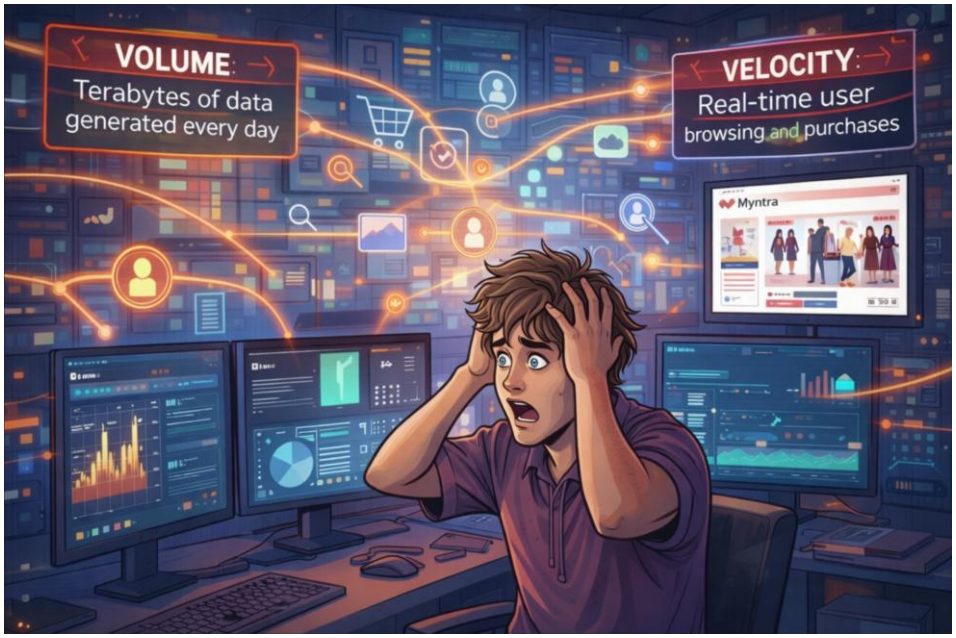
TASK 1: Big Data in Daily Life – Visual Storytelling

Chosen Application: Myntra (Online Fashion Shopping Platform)

Title: How Big Data Helps Myntra Understand Customer Fashion Choices

1. **Data Explosion**



Myntra handles millions of users browsing products, adding items to carts, searching for styles, and making purchases every day. This generates huge amounts of data continuously, showing the volume and speed challenges of Big Data.

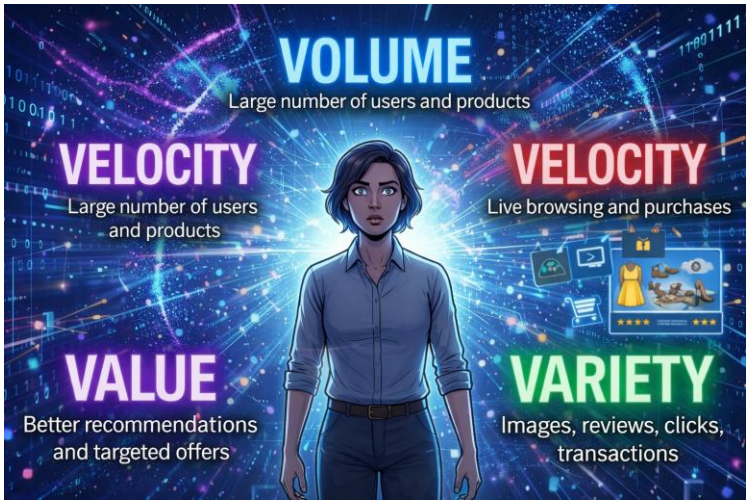
2. Variety of Data

Myntra stores different types of data such as customer profiles, purchase history, product images, reviews, and search logs. These multiple formats highlight the variety aspect of Big Data.

3. Traditional BI Limitations

Traditional BI tools mainly analyze past sales reports. They cannot process real-time user activity like trending products or sudden demand spikes during sales.

4. The 5V Model

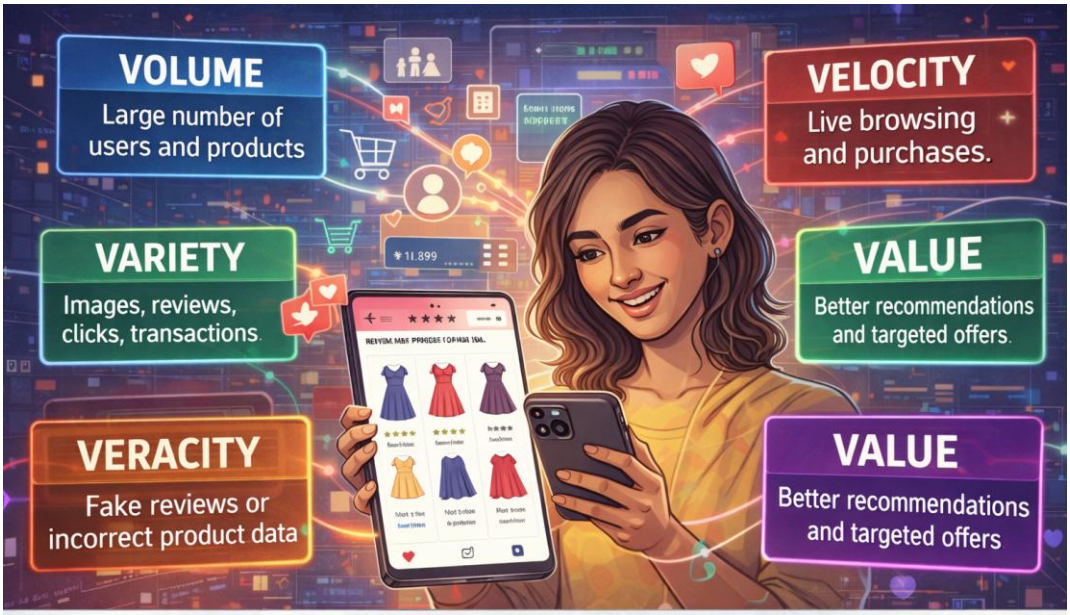


Volume: large number of users and products
Velocity: live browsing and purchases
Variety: images, reviews, clicks, transactions
Veracity: fake reviews or incorrect product data
Value: better recommendations and targeted offers

5. Big Data Solution

Myntra uses distributed storage, recommendation algorithms, and real-time analytics to track user behavior and suggest products instantly.

6. Business Value



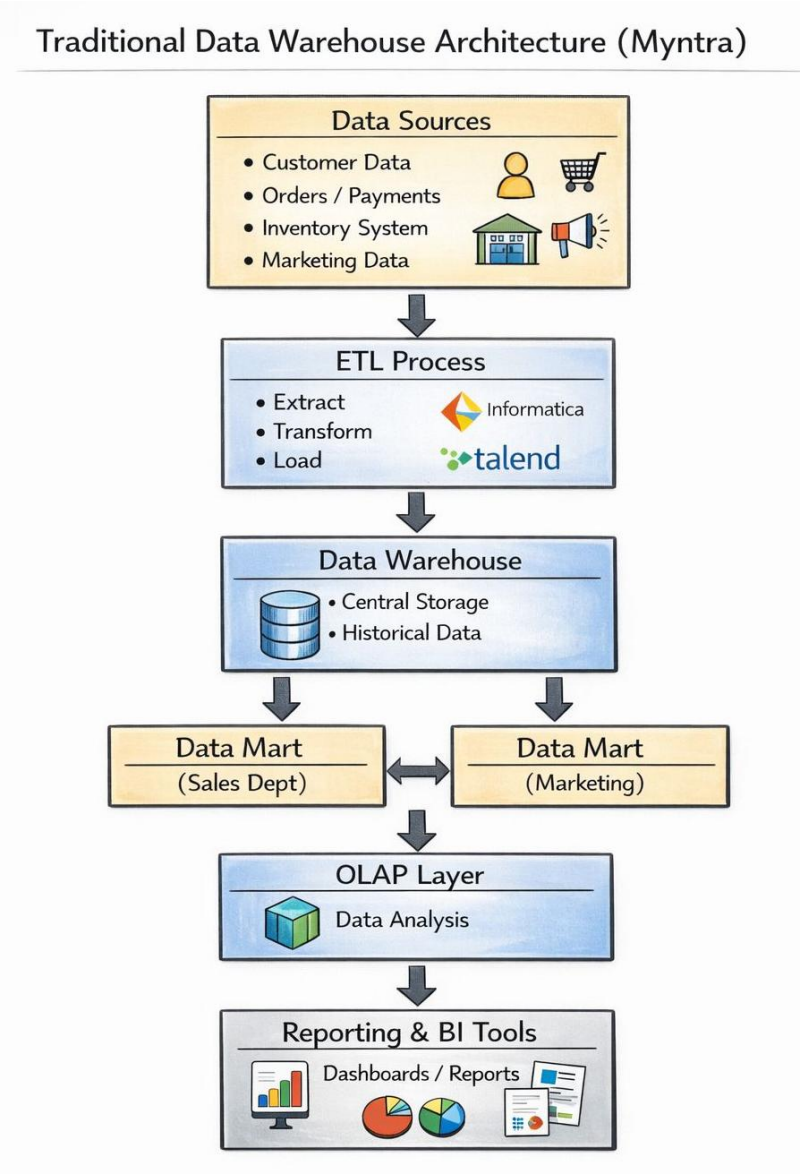
Big Data helps Myntra show personalized fashion suggestions, predict trending styles, optimize inventory, and improve customer satisfaction.

TASK 2: BI vs Big Data – Role Play

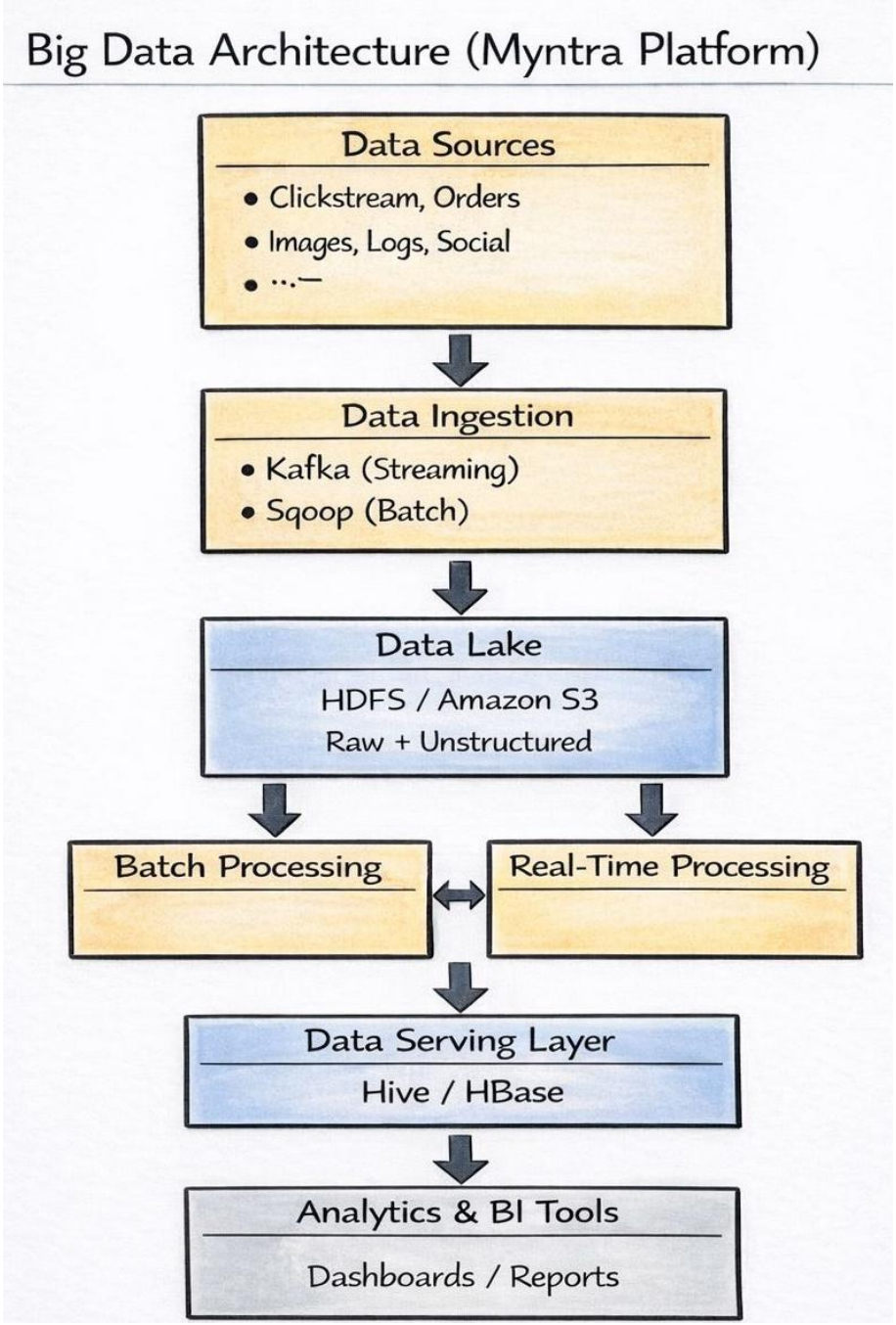
1. **Manager:** We already generate sales reports regularly. Why do we need Big Data now?
   Analyst: Sales reports show past performance, but Big Data helps us understand what customers are doing on the platform in real time.
2. **Manager:** Isn't our SQL database enough to store customer data?
   Analyst: It works well for transactions, but it becomes slow when millions of users browse, search, and click products at the same time.
3. **Manager:** What kind of extra information does Big Data analyze?
   Analyst: It tracks click patterns, search queries, time spent on products, wishlist activity, and even when users leave without buying.
4. **Manager:** So how exactly is Big Data different from BI?
   Analyst: BI mainly analyzes structured historical data, while Big Data processes large volumes of structured and unstructured data instantly.
5. **Manager:** What practical benefits does this bring to Myntra?
   Analyst: We can recommend outfits customers are more likely to buy and detect changing fashion trends early.
6. **Manager:** So this directly affects sales?
   Analyst: Yes, because customers see products relevant to their taste, which increases conversions.
7. **Manager:** Can it help with inventory planning too?
   Analyst: Definitely. Big Data helps forecast demand so we stock popular products and avoid overstocking slow sellers.
8. **Manager:** That means better decisions across departments.
   Analyst: Exactly — marketing, inventory, pricing, and customer engagement all improve with better insights.

TASK 3: Architecture Design Challenge

A. Traditional Data Warehouse Architecture (Myntra)



B. Big Data Architecture (Myntra Platform)



TASK 4: Analytics & Tool Match

Business Question	Analytics Type	Tool
How many orders were placed today and which categories sold most?	Descriptive	BI dashboards, SQL reports
Why are certain products receiving low sales or returns?	Diagnostic	Python analysis, visualization tools
Which fashion items are likely to trend in the next season?	Predictive	Machine learning models, forecasting tools
What pricing or discount strategy should be used to boost sales?	Prescriptive	AI recommendation systems, optimization tools

BONUS: Explain Big Data to a 10-Year-Old

Imagine you go to a huge shopping mall that has thousands of clothes, shoes, and accessories in different stores. Normally, it would take hours to walk around and find something you like.

Now imagine there's a super-smart helper who follows you around and remembers what you checked before — your favorite colors, your size, the brands you like, and even how much money you want to spend.

Instead of showing you everything in the mall, the helper takes you directly to the section where you are most likely to find something you love. They might also tell you when something is on discount or when new styles arrive that match your taste.

That smart helper works like Big Data.

Big Data means collecting lots of information and using it to understand people's choices better. It helps apps and websites show you useful things faster, save your time, and make your experience more enjoyable.

Github: **SAMJANADHAMALA10** **https://github.com/SAMJANADHAMALA10/Bda**